

Does Party Competition Affect Political Activism?

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Does party competition affect political activism? This paper studies the decision of party supporters to join political campaigns. We present a framework that incorporates supporters' instrumental and expressive motives and illustrates that party competition can either increase or decrease party activism. To distinguish between these competing predictions, we implemented a field experiment with a European party during a national election. In a seemingly unrelated party survey, we randomly assigned 1,417 party supporters to true information that the canvassing activity of the main competitor party was exceptionally high. Using unobtrusive, real-time data on party supporters' canvassing behavior, we find that respondents exposed to the high-competition treatment are 30% less likely to go canvassing. To investigate the causal mechanism, we leverage additional survey evidence collected two months after the campaign. Consistent with affective accounts of political activism, we show that increased competition lowered party supporters' political self-efficacy, which plausibly led them to remain inactive.

What explains why some citizens become politically active while others remain inert? The active engagement of citizens in politics is a crucial precondition for a thriving democracy. Parties play a pivotal role in this process. They allow citizens to voice their preferences and take political action. The number of party members, however, has seen substantial changes over the last two decades.¹ Party members have also become less active (Scarow 2017; Whiteley 2011; Whiteley and Seyd 2002). Meanwhile, the electoral landscape, too, has seen significant changes. In the United States, there is increasing variation in the competitiveness of local electoral races (Fraga and Hersh 2018). In

Europe, we observe a rise of new, often right-wing, parties, which has fundamentally altered the nature of party competition (Oesch and Rennwald 2018).

Can changes in electoral competition help explain variation in political engagement? More precisely, is party activism a product of competition between parties? Scarow (2017, 9) notes that “upticks in [party] membership [are] often motivated by close electoral contests” (see also Detterbeck 2005). Chong (2014, 76) seconds that “political activists relish the competition and conflict of the political arena.” Yet, empirical evidence for the link from party competition to political activism is scarce. There is a long-standing literature on the

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1. In the United Kingdom, for instance, the Conservative party has lost over 100,000 members since 2002, while Labour gained nearly 300,000 members (Whiteley et al. 2019). In Germany, the same period saw all major parties—save the Green party—lose a significant share of their members (Niedermayer 2019).

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effect of election closeness on turnout, but the findings are mixed. Focusing on the United States, some studies provide evidence for a positive competition-turnout link in elections at the federal (Grofman, Collet, and Griffin 1998; Pacheco 2008) and state levels (Flavin and Shufeldt 2015), while others find no such evidence—both for actual competition and perceived competition (Huckfeldt et al. 2007; Matsusaka 1993). The lack of clear evidence has led some to argue that greater party competition does not directly affect voters, but that it leads parties to make greater mobilization efforts (Cox and Munger 1989).

Among the few studies that focus on political engagement beyond turnout, Settle and colleagues (2016) find that residents of US battleground states are more likely to discuss politics on Facebook. Lipsitz (2009) finds that the same residents are also more likely to attend political events and to make political donations. The existing studies, however, are correlational and operate at aggregate levels, which, as pointed out by Matsusaka and Palda (1993), may give rise to aggregation bias. Most problematic, we currently lack evidence of whether electoral competition also affects party activism. The latter outcome, however, is arguably the most plausible margin along which we would expect party competition to have a meaningful effect (cf. Cox and Munger 1989). Party activism is also a highly policy-relevant outcome if we are to sustain lively political campaigns.

This article revisits the link between party competition and political activism using a field experiment. To motivate our empirical analysis, we present a simple theoretical framework that conceptualizes how increased party competition may affect the benefits a party supporter derives from becoming active. First, we consider instrumental benefits. We posit that if the supporter's party is ahead in the polls, a rise in effort by the competitor sets in motion two forces: greater competition makes a race tighter, which increases the likelihood that the supporter is pivotal and may thus spur participation (Gerber, Green, and Larimer 2008; Olson 1965). Moreover, increased competition means voters are exposed to more campaigning, which can alter the supporter's effectiveness. Second, we consider expressive benefits. We argue that on one hand, greater party competition can trigger anger toward the competitor (Huddy, Feldman, and Cassese 2007) and enthusiasm toward the supporter's own party (Marcus, Neuman, and MacKuen 2000), which may foster activism. On the other hand, greater competition may provoke anxiety (Huddy et al. 2007) and curtail the supporter's self-efficacy (Campbell, Furin, and Miller 1954; Finkel 1985), which may prevent the supporter from becoming active.

To test empirically whether party competition increases or decreases party activism, we draw on a unique empirical setting and data source. In cooperation with a major western

European party,² we implemented a field experiment during a recent national electoral campaign. Before the launch of the campaign, a survey was distributed on behalf of the party to the party's campaign email list, which was filled out by 1,417 activists. In the survey, we randomized whether respondents were given true information about the planned canvassing of members of the main competitor party. The information indicated that the effort of the competitor was exceptionally high. Given that the party we cooperated with was leading in the polls, the treatment therefore rendered the upcoming election more competitive. The control group received no information, leaving perceptions about competition unchanged.

Our main outcome is unobtrusive, real-time data on respondents' canvassing behavior in the campaign, which we obtained through a canvassing smartphone application distributed by the party. Comparing treatment to control respondents, we find that the high-competition treatment made supporters significantly less likely to go canvassing. Put differently, framing the election as more competitive decreased supporters' willingness to participate in the campaign. Treated individuals were 4.6 percentage points less likely to knock on doors during the campaign, compared to a control group mean of 16.7%. This corresponds to a 30% reduction in the likelihood of becoming active.

At what point during the campaign did the drop in participation materialize? The disaggregated temporal canvassing data shows that the effect built up continuously. It emerged a few weeks after the treatment was administered and became steadily larger as the campaign progressed. Indeed, the engagement gap between the treatment and control group was largest in the final weeks of the campaign. This temporal pattern underscores that the observed treatment effect is not simply a short-lived empirical fluke. Rather, the manipulation meaningfully shifted behavior throughout the electoral campaign.

To corroborate the fact that the treatment effect is driven by changes in supporters' beliefs about the competitor's effort, we present three pieces of evidence. First, the survey elicited respondents' beliefs about the canvassing behavior of the competitor party before and after the treatment was shown. This allows us to demonstrate that the treatment, indeed, caused respondents to update their beliefs about the competitors' effort. Second, we show that respondents who underestimate the competitor's effort—that is, those who learn that the competitor is more active than previously thought—contribute more to the observed treatment effects than overestimators. Third, we document correlationally that control group respondents'

2. We agreed to anonymize the party and the country of study.

beliefs about the effort of the competitor party are negatively associated with their own engagement in the campaign. Put differently, supporters who expect the competitor to do more canvass less.

Why did the competition treatment lead supporters to remain inactive? To parse out the underlying mechanism, we explore whether the drop in participation resulted from the fact that increased party competition reduced supporters' perceived effectiveness. Two months after the election, we therefore implemented a follow-up survey, which was filled out by 196 party supporters. Based on these data, we confirm that treated respondents exhibited significantly less confidence that their political engagement will make a difference. The treatment effect is a sizable 0.27 standard deviations (SDs) and as high as 0.34 SDs when restricting the sample to respondents who underestimated the competitor's effort level.

Besides contributing to an enduring debate about electoral competition and political engagement, our study adds to a growing literature studying the determinants of political activism in general and party activism in particular (Enos and Hersh 2015; Hager et al. 2019; Han 2016; Holbein 2017; Seyd and Whiteley 2004; Whiteley and Seyd 2002, 1998). We add to this debate by, for the first time, providing microlevel causal evidence combining unobtrusive behavioral data with exogenous variation in perceptions about party competition. Our findings demonstrate that the decision to join a campaign is decidedly affected by effort choices of political competitors: increasing perceived competition led party supporters to stay off the campaign trail. This finding contrasts with much of the existing literature, which finds either positive or zero effects of electoral competition on political activism (Huckfeldt et al. 2007; Lipsitz 2009; Settle et al. 2016; Vowles, Katz, and Stevens 2017). Moreover, a theoretical channel that we empirically explore—party competition lowering supporters' perceived effectiveness—speaks to broader evidence that links self-efficacy to political participation (Karp and Banducci 2008; Sloam 2014; Vecchione and Caprara 2009).

THEORETICAL FRAMEWORK

What explains why some party supporters join the campaign trail, while others remain inactive? This paper revisits one potential driver of campaign participation: party competition.³ In the following, we provide a theoretical framework—incorporating instrumental and expressive benefits of po-

litical activism—that helps explain how party competition affects the decision to become active in a political campaign. The theory appendix (available online) provides a formalization of our arguments.

Instrumental considerations

We begin by considering an instrumental account of party activism in which a supporter only cares about the outcome of an electoral race between two parties.⁴ The supporter may, for instance, want the party's policy goals to be implemented (Abramowitz and Saunders 2006). In such an instrumental scenario, the supporter only joins the campaign if she decides the election, that is, if she is pivotal. Since the likelihood of being pivotal is small and participation is costly (Enos and Hersh 2015, 254), scholars typically expect low levels of activism based on instrumental grounds (Gerber et al. 2008; Olson 1965). Importantly, however, the impact of volunteering in a political campaign is orders of magnitude larger than that of casting a ballot. A good canvasser convinces hundreds of people to vote. In our own sample, party supporters knocked on an average of 224 doors and roughly 40% of local races for member of parliament seats were close (within a five-point margin). Party supporters' likelihood to be pivotal—real or perceived—was thus not trivial.

Suppose further, then, that the supporter's utility gained from the election depends on her own decision to participate in the campaign as well as that of members of the competitor party. Assume that the supporter's party has a higher *ex ante* likelihood of winning the election compared to the competitor party. This was the case in our empirical setting, where the party was polling ahead of its main competitor nationally as well as subnationally. Under this scenario, increased competitor effort affects the supporter's decision calculus in two ways.

On one hand, holding constant the number of persuaded voters, an increase in competitor effort increases the likelihood that the supporter is pivotal. If the competitor steps up its game, the margin between both parties shrinks. As a result, the supporter, if active, becomes more likely to decide the election by convincing voters to turn out for her party. Put differently, the marginal impact of every voter convinced on the likelihood of winning the election increases. Following the instrumental accounts developed by Downs (1957) and Riker and Ordeshook (1968), the supporter should therefore become more likely to join the campaign when the competitor party increases its effort. A number of theoretical

3. A related literature studies the demographics, ideology, and personality of party activists (Enos and Hersh 2015; Hassell 2020; Van Haute and Gauja 2015). While this literature is important in order to understand who becomes a party activist, it does not directly speak to how external factors or strategic interactions influence political participation.

4. We use the term "supporter" to capture all individuals who lean toward a party and could plausibly join the campaign trail—be they members or not.

accounts therefore predict that as districts become more competitive, party activism rises (Beck and Heidemann 2014; Godwin and Mitchell 1982; Wolak 2006). As Pattie and Johnston (2003, 310) write, “Where one’s rivals campaign hard, there is an extra incentive to campaign harder oneself.”

On the other hand, an increase in competitor effort implies that voters are more likely to be exposed to campaigning from the opposing camp (Grofman et al. 1998). Wolak (2006, 354), for instance, notes that competitive districts see greater “campaign intensity[, which] increases opportunities for citizens to encounter campaign messages that captivate interest.” A rise in effort by the competitor may thus make it harder for the supporter, if active, to persuade or mobilize voters for her own party. Simply put, a pitch at the door may be less persuasive if the competitor follows up shortly after. The supporter’s goal is to persuade voters in favor of her party—for instance, by signaling that the party cares about the voter, by selectively emphasizing specific elements of the party’s policy platform that are of interest to the voter, and by reducing the voter’s overall uncertainty about the party’s policy positions. All three mechanisms of persuasion may be weakened if the voter is visited by a competing party shortly before or after.⁵ Increased party competition may therefore reduce an activist’s effectiveness.⁶ At the same time, it is also conceivable that activists feel more effective when party competition increases because they can counterbalance the effort of the competitor. Consequently, the effects of competition on perceived and actual effectiveness are theoretically ambiguous and ultimately an empirical question.⁷

In sum, in a purely instrumental framework, a rise in effort by the competitor affects a supporter’s decision to join the campaign in two potentially counteracting ways. On one hand, increased party competition makes a race tighter, which means the supporter’s participation is more likely to make a difference. On the other hand, increased party competition means voters are exposed to more campaigning, which might make a supporter’s engagement more or less effective. *Ex ante*, it is therefore unclear whether increased competitor effort increases or decreases party activism.

5. In a similar vein, Muller and Opp (1986) suggest that an individual’s perception about the probability that her group will succeed reinforces her sense of personal efficacy, which determines her likelihood of participation.

6. Whiteley and Seyd (1998) put forth an additional adverse instrumental channel: party supporters’ participation in a campaign may be driven by the motive to hold elected office. If the party is more likely to lose an election, this will decrease a supporter’s personal career returns.

7. In a formal model in the appendix, we show that the sign of the effect of competitor effort hinges on the “production function” of canvassing.

Expressive benefits

As emphasized by Huddy, Mason, and Aarøe (2015), a purely instrumental account of political activism is likely incomplete. When deciding whether to join a campaign, party supporters arguably also take into consideration a number of expressive benefits. Marcus and colleagues (2000) and Huddy and colleagues (2007), for example, highlight that the contextual environment in which party supporters operate can trigger differential emotions and expressive benefits, which either increase or decrease their noninstrumental motivation to take political action. How, precisely, does increased party competition affect a supporter’s emotions and expressive motives, and how does it affect her decision to take political action?

On the one hand, increased party competition may spark emotions that lead individuals to join the campaign trail. Three affective mechanisms are of particular relevance.

First, increased competition poses a threat to party supporters’ group status (Brown 2000; Tajfel et al. 1979) and potentially triggers supporters’ need to secure their party’s status and electoral dominance (Huddy, Sears, and Levy 2013; Mason 2015). A particularly important emotion in this context is anger. If a supporter feels that her group status is under threat, she may experience anger and decide to become active (Huddy et al. 2007).

Second, Glazer (2008) argues that political activism can be explained on the basis of a human desire to inflict losses on competitors. In the absence of party competition, a supporter may lack a suitable opponent and thus lack the motivation to become active. By contrast, increased competitor effort could spark a desire to see the competitor lose and thus inspire party supporters to take to the streets.

Third, political activists may simply enjoy the competitive nature of politics, where clearly identifiable groups fight for a specific goal. Greater party competition may thus spark a feeling of enthusiasm—an emotion that has been linked to higher political activism (Groenendyk and Banks 2014; Marcus et al. 2000; Valentino et al. 2011). A similar argument is made by Chong (2014, 76), who writes: “Political activists relish the competition and conflict of the political arena.”

On the other hand, increased party competition can also spark emotions that lead individuals to abstain from the campaign trail. We describe three particularly important affective mechanisms.

First, increased party competition may infringe upon supporters’ feeling of self-efficacy, which has been shown to drive political engagement (Campbell et al. 1954; Finkel 1985; Klandermans and Stekelenburg 2013). As conceptualized by Ryan and Deci (2000), self-efficacy can be curtailed when external pressures resulting from competition are present, leading to a suppression of expressive benefits (Reeve and

Deci 1996; Vallerand, Gauvin, and Halliwell 1986). As a consequence, suppressed self-efficacy and expressive benefits in response to competition may ultimately inhibit activism. Evidence for this mechanism in the political domain is provided by Whiteley and Seyd (1998), who document that political group- and self-efficacy suffer among party members after a significant election loss.

Second, greater party competition may be interpreted as a threat to a supporter's group status and can thus provoke anxiety, which has been shown to curtail political engagement (Huddy et al. 2007). As Huddy and Mason (2008, 3) write, "Threats (concerning for example a possible electoral defeat) could produce anxiety, leading to a desire to withdraw from political competition." In this regard, Settle and colleagues (2017) note that supporters with high levels of negative affectivity might be particularly likely to respond negatively to increased competition.

Third, potential activists may simply dislike the competitive nature of politics. Greater competition could thus lower enthusiasm toward their own campaign, which, as stated, has been linked to political activism (Groenendyk and Banks 2014; Valentino et al. 2011). A related phenomenon is bandwagon effects (Marsh 1985; McAllister and Studlar 1991): party supporters may be more likely to participate in a party's campaign if they perceive the party to be more likely to win the election. Greater party competition implies a higher likelihood that the supporter's party loses the election, which may lead supporters to abstain from the campaign trail.⁸

In sum, the effect of increased party competition on supporters' expressive benefits will depend on the particular noninstrumental considerations and emotions triggered. As highlighted by Settle and colleagues (2017), there is likely substantial heterogeneity across individuals that may be determined by biological factors, personality, and their general inclination to participate in political activities (Arceneaux and Nickerson 2009; MacKuen et al. 2010; Wolak and Marcus 2007). Or, as Huddy and colleagues (2015, 10) put it: "Weakly identified fans may attend games when the team is doing well and skip those where defeat is likely, but strong fans hang on and participate, even when the team is sure to lose, in order to boost their team's chances of victory."

8. In an alternative reading, bandwagon effects may not be directed at the supported party's chances of winning, but rather the total level of activism across all parties. In this case, if party competition and the total level of activism across the political spectrum are high, potential activists may be more likely to take political action, leading to a positive association between perceived competition and activists' canvassing effort.

DESIGN

Setting

To explore the impact of electoral competition on political activism, we implemented a field experiment in the context of a national election in a large western European democracy. The country of study uses a mixed electoral system. Citizens cast two votes. The first vote is for the MP of the local electoral district who must receive the plurality of votes (each party can nominate one candidate). The second vote is for a party and is proportional.⁹ Seats in the national parliament are given to all winning candidates in local electoral races. The remaining seats are allocated to preserve the proportionality of votes cast for the party (second vote).

We cooperated with one of the two main competing parties during the final months of the electoral campaign. Parties in the country of study organize their activism at three levels. At the national level, party headquarters is responsible for the overall campaign strategy and program, funding as well as providing logistical and organizational support to the grassroots campaigns. The national headquarters also takes care of most advertising (e.g., billboards, mailings, TV and radio ads). At the state level, state party headquarters have similar responsibilities at a significantly smaller scale. At the local level, party chapters are responsible for recruiting members and supporters for the campaign. Importantly, this includes recruiting and organizing the canvassing teams.

The party we cooperated with promoted canvassing as an effective campaign tool through internal communication channels as well as via the media.¹⁰ Volunteers were instructed to record canvassed doors systematically using a novel smartphone application. All of the country's well over 200 constituencies saw canvassing activity, underlining the high level of engagement. During the campaign, the party headquarters stayed in touch with local canvassers via email, social media, and telephone. One unit of the party was specifically tasked with training, supporting, and motivating local canvassers.

Survey and sample

In collaboration with the party's canvassing unit, we designed an online survey intended to collect information on current canvassing activities and potential roadblocks. The email was then sent out on behalf of the party to its list of

9. More specifically, citizens vote for the party's list of candidates for the national parliament; the list is different in each federal state and decided on by the local state's party caucus. Only parties that receive more than 5% of votes are taken into consideration.

10. Although door-to-door canvassing has been a widely used campaign strategy in many countries, it had been used comparatively less frequently in the country of study and by the party we cooperated with.

supporters who had expressed interest in supporting the party. In the email, the supporters were asked to participate in a survey and informed that the answers would be used to help organize the party's canvassing activities. The invitation email was designed by the party and used a standard party template. A reminder email was sent one week after the initial invitation. Overall, 1,417 party supporters agreed to participate. The response rate was 5.6% among all sent emails and 16.2% among all opened emails. Importantly, participants were not aware that the data might also be used for scientific purposes.¹¹

The survey was brief in nature so as to minimize dropout, including just ten questions. The survey instrument is provided in the appendix. The first six items measured background information. As summarized in table A1 (tables A1–A21 are available online), we inquired about respondents' gender (23% female), age (mean of 40), party membership (mean of 82%), years of membership (mean of 12), prior canvassing experience (mean of 37%), and attendance at campaign workshops (mean of 22%). In addition, we know whether respondents had downloaded the canvassing smartphone application (discussed below) before the survey (28%).

How does the supporter sample compare to the party's full population of members? In order to maintain anonymity, we cannot provide precise figures. Broadly speaking, the sample matches the party's distribution of members regarding gender and geography. However, the sample is significantly younger than the average party member. Our sample is also disproportionately more engaged. It includes 12.9% of all party supporters who canvassed for the party during the entire campaign. Furthermore, survey respondents were responsible for 21.9% of all knocked doors during the campaign (as measured with the smartphone app). The sample can thus best be characterized as "young and highly motivated supporters." This group is relevant because it includes individuals for whom the party could have hoped to increase engagement. Given the mild nature of the intervention and the relatively high-effort nature of canvassing, the sample characteristic increases our ability to detect treatment effects. Moreover, the young age in our sample also implies that supporters did not face technological barriers to using the smartphone application with which the party organized its canvassing and which we use to obtain unobtrusive behavioral outcomes.

Still, we must caution that the behavior of our sample cannot be generalized to all party members. Our study merely allows us to draw internally valid conclusions about the strategic behavior of the party's active supporters who were interested in joining the campaign and who took the survey. This sample, however, is highly relevant. In our setting, the majority of canvassing (and campaigning at large) was done by a relatively small subset of motivated party supporters—individuals that took part in our survey.

Treatment

To study the effect of party competition on supporters' decision to participate in the campaign, we administered an information treatment to shift exogenously participants' beliefs about the canvassing effort of members of their main competitor party. Figure A.1 (figs. A.1–A.5 are available online) summarizes the experimental design, which consisted of three steps.

First, we elicited all respondents' pretreatment beliefs about the canvassing intentions of the competitor party. Specifically, we asked, "Think of 100 typical [competitor name] party members. What do you think: How many of these 100 [competitor name] party members plan to go canvassing in this electoral campaign?" On average, as table A1 documents, respondents believed that 21% of members of the competitor party planned to go canvassing.

Second, we randomly assigned half of the participants to receive information about the canvassing intentions of members of their main competitor party. Respondents in the control group received no information.¹² Participants in the treatment group were shown the following treatment information that displayed the previously stated belief of the respondent and compared it to the true figure: "You said [X] out of 100 [competitor name] party members. According to a survey of [competitor name] party members, 56 out of 100 [competitor name] party members plan to go canvassing in this electoral campaign."¹³

We gathered the information on canvassing intentions of the major competitor party in a different survey, which we had administered six weeks before the main experiment. This statistic was perceived as exceptionally high, given that 89% of respondents underestimated this number (see fig. A.3). Importantly, as the party we cooperated with was

11. To our knowledge, the party did not mention the scientific cooperation for two reasons. First, the email constituted regular contact with canvassers and thus resembled "business as usual." Second, the scientific cooperation with the party was subject to an agreement specifying that the setting would be kept confidential. We discuss the ethical considerations of our study in the section on ethical considerations.

12. Table A2 shows that randomization achieved excellent balance for the aforementioned covariates across the treatment and control group. A joint *F*-test from a regression of the treatment indicator on all available preregistered covariates is insignificant ($p = .471$).

13. The treatment screen also displayed the two numbers using a bar chart (depicted in fig. A.2). This served the purpose of making the two numbers more readily accessible and comparable.

leading in the polls, the treatment information rendered the race more competitive than previously thought.

Third, we elicited all respondents' posttreatment beliefs about the actual canvassing effort of the competitor. This allows us to assess whether treated respondents updated their posttreatment beliefs. We inquired about "actual" behavior because we hypothesized that actual behavior is ultimately the relevant metric that affects the electoral outcome. Specifically, we asked, "What do you think: How many of these 100 [competitor name] party members will actually go canvassing during this electoral campaign?" After eliciting respondents' beliefs about canvassing intentions, providing the treatment, and eliciting the posttreatment beliefs about actual behavior, we administered two attitudinal outcome measures. First, respondents were asked whether they planned to go canvassing in the campaign (mean of 53%; see table A1). Second, respondents were asked how many days they planned to go canvassing (mean of 4.1; see table A1).

Outcome: Real-time canvassing behavior

To measure respondents' canvassing behavior unobtrusively, we draw on unique real-time data on actual canvassing activity collected by the party during the campaign. The party provided a novel smartphone application to its supporters to record knocked doors. The party instructed and encouraged all canvassers to download and use the app. Frequent reminders via social media and email ensured that many active canvassers used the app. The party also provided the canvassers with training workshops in which party supporters were encouraged to use the app. To further boost take-up, the app included a number of gamification elements. Before the survey took place, 28% of the sample (400 supporters) had already downloaded the app. Another 80 supporters downloaded the app after the experiment. The unobtrusive, geo-coded app data therefore provide a unique lens into the actual canvassing activities of respondents.¹⁴

The app data allows us to construct three behavioral outcomes: First, we construct a dummy for whether respondents engaged in any canvassing in the time from the survey until the election (average of 14%; see table A1).¹⁵ This measure captures respondents' participation decision—that is, whether to canvass or not. By contrast, our second and third measures

capture respondents' effort decision conditional on participating in the campaign. The second measure involves the number of days respondents went canvassing (average of 0.65 days; see table A1). As our third outcome, we measure the total number of doors respondents knocked on, winsorized at the 99th percentile (average of 32 doors; see table A1).¹⁶

RESULTS

Empirical specification

To assess the impact of the information treatment, we estimate the following preregistered specification using ordinary least squares:¹⁷

$$\text{Canvassing}_i = \pi_0 + \pi_1 \text{Treatment}_i + \zeta^T X_i + \epsilon_i,$$

where Canvassing_i is one of the measures of respondent i 's party activism. Treatment_i is a dummy variable taking a value of 1 for people who receive the high-competition information and 0 otherwise. And X_i is the preregistered set of control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in a campaign workshop, whether a participant has already downloaded the canvassing smartphone application, whether a participant has participated in canvassing before this national election, and whether a participant has already canvassed during this election (see table A1 for an overview). We report robust standard errors in all models.

Effect on canvassing behavior

Did the high-competition treatment affect respondents' canvassing behavior? We begin by analyzing respondents' participation decision. Columns 1 and 2 in table 1 show that the treatment led to a significant decrease in respondents' party

14. Reassuringly, the sample of canvassers is fairly homogeneously spread out over the country of study. In our sample of survey respondents, we observe canvassing in 81.4% of all constituencies. Furthermore, canvassing activity is evenly spread across constituencies, with no constituency containing more than 1.5% of our sample.

15. Following our preanalysis plan, individuals who do not appear in the canvassing application data are coded as 0. We show that all results are robust to a different coding scheme in the section titled "Robustness."

16. One may be worried that the smartphone app affected respondents' canvassing behavior. We cannot, unfortunately, observe the behavior of respondents who did not download the app. Fortunately, the party strongly encouraged everyone to use the app. The behavior recorded in the app thus constitutes how a motivated, regular canvasser behaved. We should also reiterate that canvassers were not aware that researchers would look at the data. This arguably diminishes concerns about potential "observer effects" (see a more elaborate discussion in the Ethics section). Reassuringly, in table A4, we show that control group respondents who used the app were 134% more likely to report an intention to go canvassing. Our sample thus likely captures the highly active canvassers. That said, we do not know if supporters' canvassing behavior would have been different had the party not used the app. Given that similar apps are used by many parties across Europe and the United States, however, our evidence arguably captures regular canvassing behavior in today's political campaigns.

17. Our results are robust to using logit regressions instead of ordinary least squares. Results are shown in table A6.

Table 1. Effect of High-Competition Treatment on Canvassing Behavior

	Any Canvassing		Days Canvassed		Doors Knocked	
	Without Controls (1)	With Controls (2)	Without Controls (3)	With Controls (4)	Without Controls (5)	With Controls (6)
Treatment	-.046** (.018) [.013]	-.032** (.015) [.036]	-.117 (.146) [.422]	-.044 (.135) [.751]	-3.623 (7.532) [.632]	.024 (7.042) [.998]
Control mean	.160	.160	.712	.712	34.148	34.148
Controls	No	Yes	No	Yes	No	Yes

Note. $N = 1,417$. Coefficients, robust standard errors (in parentheses), and randomization inference based p -values (in brackets) of a linear model (ordinary least squares) regressing the indicated behavioral outcomes on the treatment. Outcome data are obtained through a smartphone application used by the canvassers to register knocked doors. “Any Canvassing” refers to any recorded canvassing activity. “Days Canvassed” refers to the number of canvassed days. “Doors Knocked” refers to the number of knocked doors winsorized at the 99th percentile. Odd columns display specifications without control variables. Even columns include the following prespecified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online app, whether a participant has participated in canvassing before this national election, and whether a participant has canvassed in the current election.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

activism. In the combined sample—that is, pooling under- and overestimators—we find that treated respondents are 4.6 percentage points less likely to participate in the campaign. This corresponds to a 30% decrease in the likelihood of becoming active. When including the preregistered control variables, we estimate a reduction of 3.2 percentage points. Both estimates are significant: the randomization inference p -value demonstrates that, when considering all possible treatment assignments and assuming no effect for any individual, the probability of observing a reduction in engagement of this magnitude (by chance) is just 1%.¹⁸ These findings indicate that the treatment had a significant negative effect on respondents’ participation decision—that is, whether to canvass or not.

Next, we investigate respondents’ effort decision—that is, how many days to canvass and how many doors to knock. Here, the coefficients are noisily measured. Three of the four reported coefficients imply that the treatment reduced respondents’ effort decision. When excluding controls (cols. 3 and 5 in table 1), for instance, treated individuals seemingly canvassed 0.12 fewer days (–16.4%) and knocked on 3.6 fewer doors (–10.6%). But the estimates are not significant. In ad-

dition, when including control variables (cols. 4 and 6 in table 1), the estimates are very close to zero and not consistently negative. Our evidence for respondents’ effort decision is thus inconclusive.

Taken together, the preregistered models yield robust evidence that learning about high activism in the competitor party had a negative effect on individuals’ participation decision. We find no effect for individuals’ effort decision. These effects suggest that “marginal” activists who would have exerted relatively little canvassing effort left the campaign.¹⁹

Dynamics of behavioral change

At what point and for how long did the treatment change respondents’ engagement? One concern with experimental studies is short-lived treatment effects. The real-time behavioral data at our disposal allows us to address this concern. In figure 1, we plot the cumulative distribution of canvassing participation over the course of the campaign as well as 1 standard error intervals (*short vertical lines*). The dotted line plots the treatment group, and the solid line plots the control group. The treatment was administered in week 0 (*long vertical line*), while the election took place in week 8.

The figure provides three pieces of evidence. First, treatment and control respondents are indistinguishable in the weeks preceding the experiment, underscoring the fact that

18. As an additional robustness check, using the sample of participants who downloaded the canvassing application and for whom we know their location, table A5 controls for local political circumstances by including a dummy variable that indicates whether the vote share difference between the party and its strongest competitor is within 5 percentage points. Results are virtually unchanged.

19. For the sake of brevity, we report preregistered heterogeneity analyses in the heterogeneity section of the appendix.

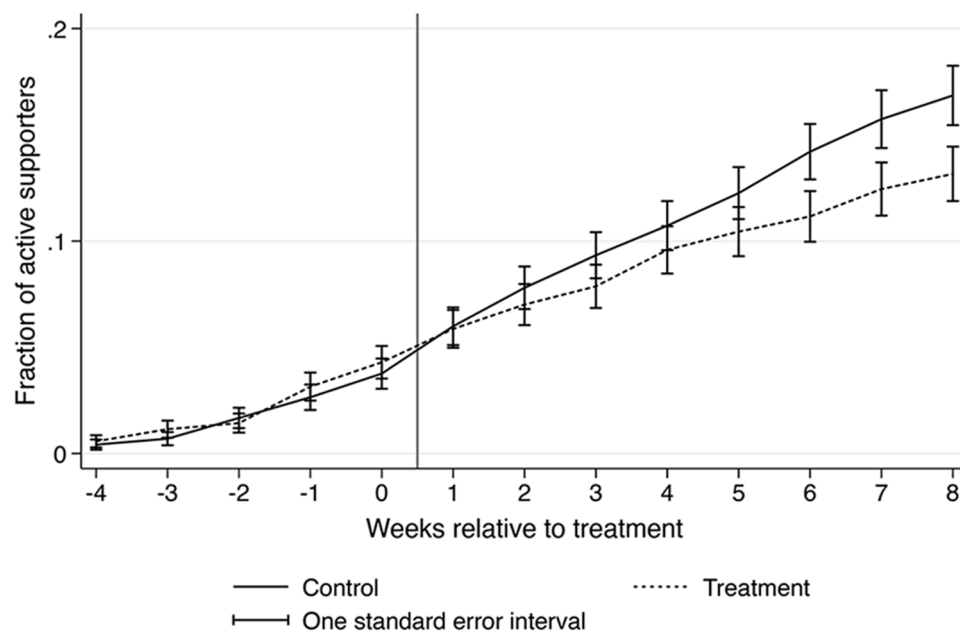


Figure 1. Participation in campaign over time across the treatment and control group. Raw fraction of respondents who participated in the canvassing campaign at any point for the treatment group (dotted line) and control group (solid line). Short vertical lines are 1 standard error intervals. The long vertical line indicates the timing of the treatment. The election took place in week 8. No control variables are included.

differences between the treatment and control group in the weeks following the experimental manipulation are caused by the treatment. Second, there is a continuous buildup of engagement as the election day approaches. Third and most important, the treatment group—starting a few weeks after the randomized information provision—shows significantly lower levels of engagement. Furthermore, the gap between both groups widens at a roughly constant rate as the campaign progresses.²⁰ The continuous buildup of the effect thus underscores that the treatment had a pronounced and persistent effect on respondents' party activism.²¹ The temporal pattern also holds when controlling for time-invariant individual characteristics (see fig. A5).

Robustness

Before turning to mechanisms, we provide two robustness tests regarding our outcome measure. First, following our preanalysis plan, we have coded respondents who did not use

the app as not having canvassed.²² If some of these respondents did canvass, we classify them incorrectly and thus add measurement error to our outcome variable. This measurement error, however, is not problematic, given that imputing zero canvassing for individuals without the application should attenuate the estimated treatment effect. The reported treatment effect thus represents a lower bound. Still, to further mitigate concerns regarding this imputation, table A5 presents our main model, excluding respondents who did not download the app or who did not register a valid location. Reassuringly, the results confirm our treatment effect, which, if anything, becomes larger.

Second, one may be worried that the treatment affected uptake of the app differentially across the treatment and control group. Reassuringly, a large fraction of party supporters had already downloaded the app before the survey was fielded. This makes it unlikely that the treatment changed app download behavior. We confirm this conjecture in table A3. Specifically, we find a zero effect on individuals' likelihood to click on a link in the survey that took respondents to the App/Android store to download the app. Furthermore, we also find no evidence of differential app download between the treatment and control group after the survey (not for the

20. We attribute the lack of significant differences early on in the campaign to reduced power resulting from lower levels of canvassing activity.

21. Two factors may contribute to the persistent effect of our information treatment. First, the information was provided through official party channels, which likely made the information salient and credible and thus produced a long-lasting impression on participants. Second, the information about canvassing intentions of the competitor party was 163% above what the average supporter believed. The impact of the treatment on respondents' beliefs was thus large, which helps explain its lasting effect.

22. NB: we preregistered and use the full survey sample—rather than the app user sample only—in order to avoid posttreatment bias, which would arise if the treatment had affected app downloads.

24-hour time window after the survey, one week after the survey, or any time after the survey).²³

MECHANISMS

Why did the high-competition treatment reduce party activism? In this section, we explore two mechanisms. First, we probe a more immediate mechanism, namely, whether the treatment reduced engagement by changing respondents' beliefs about the competitor party's effort. Second, we revisit the proposed affective mechanism whereby increased party competition reduces activism by dampening respondents' perceived effectiveness (Ryan and Deci 2000).

Beliefs about competitor effort

Did the treatment change respondents' behavior by changing their beliefs about the competitor party's effort? We provide three pieces of evidence in favor of this channel.

First, the treatment did indeed shift respondents' post-treatment beliefs about competing canvassers' behavior. Figure A.4 shows the difference between post- and pretreatment beliefs across all participants in the control and treatment group. We observe that underestimators in the treatment group update positively relative to the control group underestimators. The reverse holds for overestimators (a formal test is provided in table A7).

Second, as a qualification for beliefs as a causal channel, we would expect underestimators to drive the treatment effects. Put differently, relative to overestimators, underestimators should canvass less as they learn about greater effort by the competitor. To test this empirically, in table A8, we interact the treatment with a dummy for overestimation. Unfortunately, the sample includes very few overestimators (10%). We therefore lack the statistical power to estimate reliably the differences between the two groups. With that said, table A8 does show a positive interaction between the treatment and overestimation, suggesting that the results are driven by underestimators. However, because of the lack of statistical power, the estimate should be interpreted with caution.

23. As a further piece of robustness, we discuss the possibility of demand effects in the Demand Effects section of the appendix. Specifically, we investigate the effect of our treatment on two self-reported measures of respondents' intentions to go canvassing. We do not find a significant negative treatment effect on these self-reported outcomes. If anything, there is a slight positive effect (though the coefficient is insignificant without control variables). This short-term effect may be a product of social pressure induced by the surveyor, as respondents might want to please the party by showing a higher willingness to engage in the campaign. The inconclusive finding thus, if anything, underscores the methodological necessity of unobtrusively collecting behavioral data.

A third way to assess the importance of beliefs about competitor effort—though not causally identified—is to correlate control respondents' posterior belief about competitor effort with their canvassing behavior. Table A13 demonstrates that control respondents' beliefs correlate negatively with their canvassing behavior. Put simply, individuals expecting the competitor to do more canvass less. The correlation is significant for all three behavioral measures. Moreover, the models are robust to the inclusion of the preregistered control variables. The analysis thus lends further empirical support that beliefs are a plausible channel through which the treatment affected respondents' political activism.

Reduced perceived effectiveness

Did the treatment reduce respondents' activism by making canvassers feel less effective? To test this mechanism, we investigate whether the treatment, indeed, changed activists' perceptions about their effectiveness. To measure perceptions about campaign effectiveness, a follow-up survey was conducted with the same pool of party supporters two months after the election. We reached 196 respondents. While this number is low, there is no evidence that there was differential selection into the second survey. First, the original treatment status does not predict participation in the postelection survey (tables A9, A10). Second, the treatment and control groups in the postelection survey are balanced on predetermined covariates (tables A11, A12).

To measure whether treated respondents felt less effective, we asked to what extent they agreed with the following statement: "You can change political circumstances by becoming politically active." Answers were scored on a seven-point Likert scale from "Do not agree at all" to "Fully agree." We standardize this variable to have a mean of 0 and an SD of 1. Table 2 shows that treated respondents show a significantly lower level of perceived effectiveness. The treatment effect is a substantively sizable 0.25 SDs. When we restrict the sample to underestimators—respondents who initially underestimated the competitor's effort level and learned that the competition will do more—the estimated effect size increases to 0.31 SDs. The large reduction in activists' perceptions about their ability to drive political change, detected two months after the election, provides suggestive evidence that increased party competition, indeed, lowered activists' perceived effectiveness and thereby dampened engagement.

Substitution to other effort domains

We have provided robust evidence that increased party competition leads the supporters of a political party to shy away from canvassing. This evidence raises an important follow-up question: Did party supporters abstain from the campaign

Table 2. Effect of High-Competition Treatment on Perceived Effectiveness (Ability to Make Political Change)

	Pooled Sample		Underestimators	
	Without Controls (1)	With Controls (2)	Without Controls (3)	With Controls (4)
Treatment	-.273* (.143)	-.245* (.145)	-.339** (.149)	-.314** (.151)
Control mean	.131	.131	.150	.150
Observations	196	196	180	180
Controls	No	Yes	No	Yes

Note. Coefficients and robust standard errors (in parentheses) of a linear model (ordinary least squares) regressing our measure of perceived effectiveness on the treatment for individuals who responded to the post-election survey. Columns 1 and 2 present the estimates for the full sample. Columns 3 and 4 present estimates for underestimators. Perceived effectiveness is measured using respondents' agreement to the statement: "You can change political circumstances by becoming politically active." Answers were scored on a seven-point Likert scale ranging from "Do not agree at all" to "Fully agree." The outcome is standardized. Prespecified control variables include party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election, and whether a participant has canvassed in the current election.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

entirely or did they merely shift their effort to other modes of campaigning? Canvassing is one of the most effective forms of campaigning (Green, McGrath, and Aronow 2013). However, perhaps a rise in competitor effort on the streets leads party supporters to choose new channels to contribute to the campaign? This may be particularly relevant if canvassers want to avoid fierce competition.

To test for this substitution effect, we use a second feature of the canvassing app. In particular, the app also gave party supporters an opportunity to share party news stories on Facebook. As shown in tables A14 and A15, we find small and insignificant treatment effects on activism on social media. Treated respondents were no more likely to share stories about the party on Facebook. The evidence thus implies that treated individuals did not shift their effort toward other forms of campaigning—in this case, social media. Rather, they abstained from any engagement.

ETHICAL CONSIDERATIONS

Field experiments are an excellent method for drawing causal inferences. However, they also raise tough ethical questions

because researchers intervene in (rather than observe) the real world. In our case, ethical considerations were particularly pressing because our study could have had an impact on the election. We therefore carefully considered the ethical dimension of our study, which we want to discuss before concluding. While we obtained ethical approval at the University of Oxford, we still want to reflect on two particular ethical issues: potential effects on the election and subjects' noninformation about participation in an experiment.

First, implementing the survey meant that we intervened in an electoral campaign. Were we justified in doing so? Importantly, the survey among party supporters would have taken place with or without our presence. The party regularly engaged its supporters using emails, surveys, and phone calls. We simply advised the party on how to implement the survey. The ultimate decision to launch the survey, however, was made by party officials. There was also no power differential that could have led the party to feel obligated to implement the survey. At the time, all authors were graduate students and the party is one of Europe's largest, with a highly professional team of campaigners.

Second, and related, the expected sample size meant that it was exceedingly unlikely for the study to have any effect on the election. Today, we know that this calculation was correct. Controlling for prespecified covariates, we do not find any aggregate impact on doors (table 1, col. 6). Even if we take the results without controls, the treatment group knocked on 3.6 fewer doors. Assuming a persuasion rate of 10% based on Pons (2018), this means that the experiment lowered the party's number of votes by 258. This means not a single constituency would have elected a different candidate had the study not taken place. The national share of the party was also entirely unaffected (precise to the fourth decimal). All this is not to say that the survey was without any effect. We did, after all, intervene in the real world. However, it strikes us that the scientific insights—presented above—were sufficiently high to justify our intervention.

Third, the survey did not deceive subjects. Party supporters were provided with truthful information about the effort of the main competitor. Study participants—who were contacted online—were also entirely free in their decision to participate in the study. The party did not, however, inform subjects that the data would also be used for scientific purposes. This noninformation worked in our favor by preserving the natural field setting, in which "the environment is one where the subjects . . . do not know that they are in an experiment" (Harrison and List 2004, 1014). That said, we hope that (a) by avoiding any harm; (b) by allowing subjects to freely choose to participate; and (c) by maintaining the confidentiality of all subjects, including the party and country,

we were justified in our acceptance of this noninformation in order to explore an important question in political science.

CONCLUSION

This paper has provided evidence that increased party competition makes party supporters less likely to join the campaign trail. We conducted a field experiment with a major European party during a recent national electoral campaign. In the experiment, party supporters were randomly assigned to true information about the canvassing intentions of members of the major competitor party. The information indicated that the canvassing effort by the competitor party was exceptionally high. Given that the party we cooperated with was leading in the polls, the treatment therefore increased the perceived competitiveness of the upcoming election. Using unobtrusive, real-time data on respondents' canvassing behavior, we found that treated respondents were 3–5 percentage points less likely to go canvassing. We also provide suggestive evidence as to why activists reduced their engagement: increased competition meant that activists felt less effective.

A natural follow-up question is whether our main finding generalizes to the broader population (Van Haute and Gauja 2015). To address generalizability, we compared our sample to a general population panel survey in the country of study. Relative to an equal split of men and women in the general population (49% men), our sample is predominantly male, with only 23% female. With an average age of 40, our sample is also much younger relative to the average age of the population. Finally, we also note that our sample's respondents below the median age are 0.1 SD more likely to believe that social and political activism influences societal conditions. Given that men in the general population are 5 percentage points more likely to support a party and also exhibit a significantly higher interest in politics (+0.39 SD), our sample thus likely captures highly active and interested party supporters. Taken together, these statistics indicate that a general population sample might likely be less responsive to changes in political competition.

A second and related question is whether our main outcome—engagement in a canvassing campaign—generalizes to other forms of political activism as well as to different types of electoral systems. Reassuringly, we found that there are no substitution effects to online campaigning. This makes it plausible that the treatment led potential supporters to abstain from any campaign engagement. We cannot answer, however, the question of whether the treatment would have worked differently in other electoral systems. Fortunately, the country of study uses a mixed electoral system. There is thus hope that the results apply under both proportional and

majoritarian representation. However, further evidence is necessary to settle this question.

With these questions in mind, we want to conclude our study with a word on normative implications. If political activists reduce their engagement when competition increases, what does this mean for the democratic political process? If taken at face value, our finding implies that fierce competition may not help spur democratic engagement. Perhaps citizens prefer a more consensual, communicative political process (Habermas 1985). Given that politics is not, or at least should not be, a zero-sum game, excessive campaign effort by one party may thus do more harm than good in terms of fostering political engagement.

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